Newton's Ring

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Aim of the Experiment

To study the interference pattern and determine the radius of curvature of the planoconvex lens using Newton's rings apparatus.

Apparatus Required

(i) Source of light (Hg lamp)

(ii) Newton's ring apparatus

(iii) Power supply for Hg CS/50 W lamp

(iv) Interference filters (Green & yellow)

(v) Mounted lens & Screen

Figure-1: Experimental set up for Newton's ring.

Theory:

In Newton's rings experiment, a parallel beam of monochromatic light is incident normally on a combination of a planoconvex lens L and a glass plate G as shown in Fig.1. A part of each incident ray is reflected from the lower surface of the lens. The other part after refracting through the air film between the lens and the glass plate is reflected back from the glass plate. These two reflected rays from coherent sources of light. As a result, they interfere and produce a system of alternate dark and bright rings with the point of contact between the lens and the plate as the centre. These rings are known as Newton's rings. Looking at the diagram we can write,

$r_m^2 = R^2 - (R - t)^2$ or $2t = r_m^2 / R$ When $2Rt >> t^2$ or $2R >> t$

Where r_m is the radius of the mth bright fringe,

R is the radius of the curvature of the planoconvex lens,

t is the thickness of the air film .

For bright fringes $2t\mu \cos\theta = m\lambda$

Where μ= refractive index of the medium

θ= refraction angle

λ=wavelength of the monochromatic source

m= order of the fringe

As we know, μ=1 for air and θ to be very small, we have

$$2t = m\lambda \quad \text{or} \quad r_m^2 / R = m\lambda$$

From this relationship one can obtain the wavelength of light if the radius of curvature of the plano-convex lens is known.

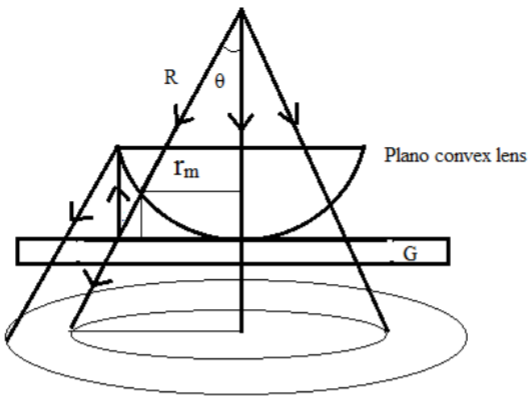
From the above expression we can write $D_m^2 = 4r_m^2 = 4Rm\lambda$.

If D_n is the diameter of the $(n)^{th}$ bright ring then $D^2_n = 4Rn$.

So we can write,

$$R = \frac{D_n^2 - D_m^2}{4(n-m)\lambda} \text{ for } n>m$$

From this the value of "***R***" the radius of curvature of the planoconvex lens can be determined.



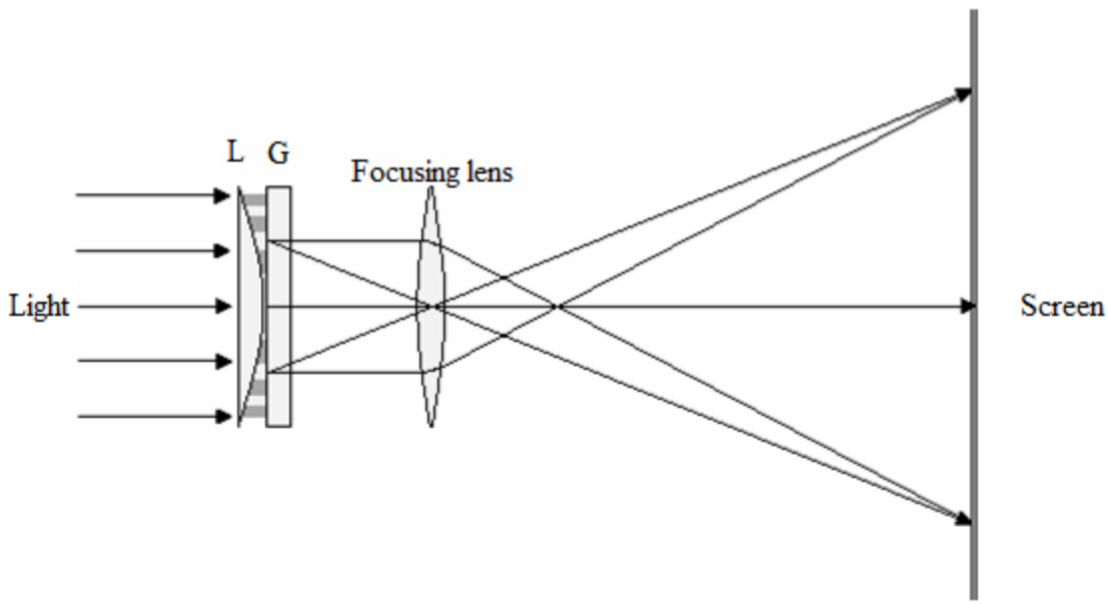


Figure-2: Schematic diagram of Newton's ring.

Procedure:

(i) All the requisite apparatuses including the mercury spectral lamp is mounted on the optical bench.

(ii) Switch on the universal choke of the power supply and wait for the warm-up for a few minutes.

(iii) Adjust the position of lens and Newton's ring apparatus to obtain coloured rings on the screen.

(iv) Clamp the green filter in the object holder.

(v) Measure the diameter of mth and n^{th} ring on the screen.

(vi) Repeat the measurement with blue colour filter.

(vii) Plot a graph between radii of the ring as a function of order number for various wavelengths.

Observation:

Table-1

Sl. No.	Colour of the filter & its wavelength	Diameter of nth ring (D_n)(in mm)	Diameter of $m=1^{th}$ ring (D_m) (in mm)	$D_n^2 - D_m^2$	R(m)
1	YELLOW **(578 nm)**	D_1=9.4		0	
2		D_2=11.6	D_1=9.4	46.20	19.98
3		D_3=13.4		91.20	19.72
4		D_4=15.0		136.64	19.70
5		D_5=16.0		167.64	18.13
6		D_6=17.4		214.40	18.55
7		D_7=18.4		250.20	18.04
8		D_8=19.4		288.00	17.80
9		D_9=20.6		336.00	18.17
10		D_{10}=21.4		369.60	17.76

Table-2

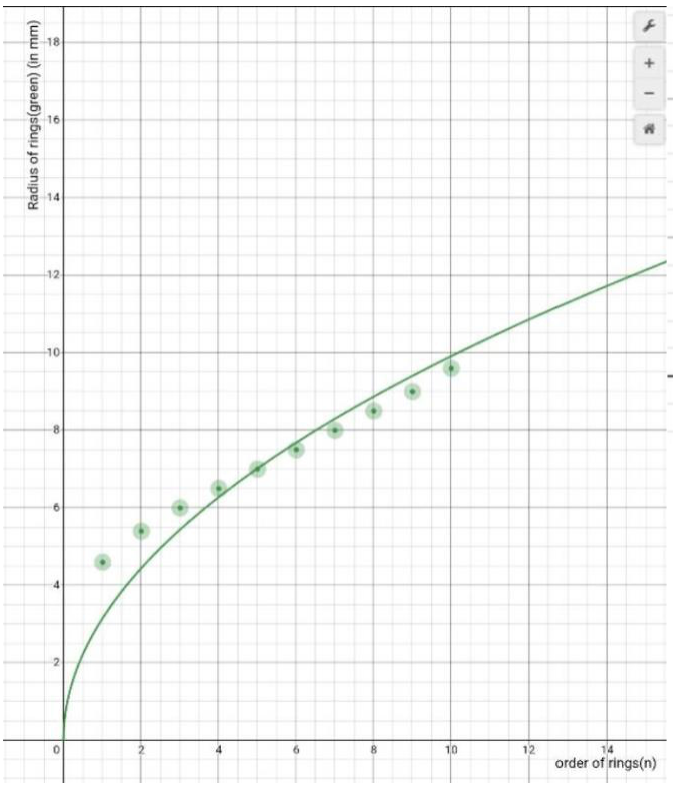
Sl. No.	Colour of the filter & its wavelength	Diameter of nth ring (D_n)(in mm.)	Diameter of $m=1^{th}$ ring (D_m)(in mm.)	$D_n^2 - D_m^2$	R (m)
1	GREEN **(546 nm)**	D_1=9.2			
2		D_2=10.8	D_1=9.2	32.00	14.65
3		D_3=12		59.36	13.59
4		D_4=13		84.36	12.88
5		D_5=14		111.36	12.75
6		D_6=15		140.36	12.85
7		D_7=16		171.36	13.07
8		D_8=17		204.36	13.37
9		D_9=18		239.36	13.70
10		D_{10}=19.2		284.00	14.45

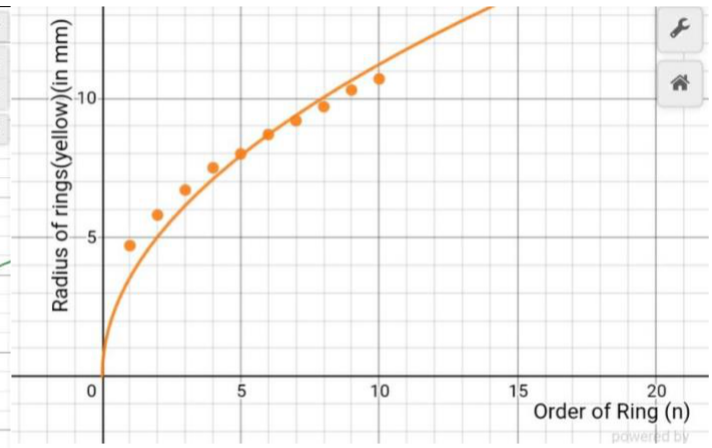
Table-3

Sl. No.	Radius of ring	Green	Yellow
1	**r_1**	4.6	4.716981
2	**r_2**	5.4	5.849057
3	**r_3**	6	6.698113
4	**r_4**	6.5	7.45283
5	**r_5**	7	8.018868

6	**r_6**	7.5	8.679245
7	**r_7**	8	9.245283
8	**r_8**	8.5	9.716981
9	**r_9**	9	10.28302
10	**r_{10}**	9.6	10.66038

Graphs:





Calculation:

The radius of curvature obtained from data of yellow filter= 18.65m ,σ= 0.84m

The radius of curvature obtained from data of green filter= 13.48m ,σ= 0.65m

Average radius of curvature= (16.06 +/- 0.74) m.

Precautions

(i) One should carefully handle the Newton's ring set up while moving the mounted screws.

Considerations

(i) What is interference of light and what are the different ways in which light interferes?

Ans: Interference can be defined as the superposition of light coming from various close (separation of the order of wavelength) monochromatic coherent sources. Due to the wave character of light, the light may interfere constructively (constructive interference) and the amplitude/brightness of a region is greater than its surrounding regions. Or the light may interfere destructively and cause the brightness of certain regions to be low.

(ii) How interference takes place in case of a Newton's ring?

Ans: The interference pattern in case of a newton's ring is formed by the superposition of light going in two paths. One reflecting off of air-lens interface and the other refracting and then reflecting off of the glass plate.

(iii) Why the fringe pattern obtained from Newton's ring is circular and can it be straight?

Ans: The fringe pattern thus formed is circular due to the shape of the air film which in turn has constant thickness around a circle so that the path difference of light for the entire circle is same. Hence the fringes formed are circular. To obtain straight fringes, we may use straight thin wedges or prisms with small refracting angles.

Conclusion:

(i) Interference pattern due to thin film of material(air) is observed. Wave nature of light is confirmed. From the experimental data obtained, the radius of curvature of the lens was calculated. The graph of radii of rings as a function of order of rings was plotted and it vaguely resembled a parabola.

(ii) Some limitations faced by this experiment are need of monochromatic light of suitable wavelength: for instance, the experiment would not work for normal light because of formation of multiple colored rings whose radii cannot me measured accurately. Also, adjusting the lens-plate gap for obtaining a clean interference pattern is very hard. Radii of rings observed is actually not totally according to the theory as our assumptions ($\cos\theta \sim 1$ and thin film approximation) fail for higher order fringes and distortion of rings becomes more and more as we travel farther from the center.

(iii) Although the experiment (as described here) does not find many real world applications (apart from a proof of concept), the concepts which come into play here have tremendous applications. Interference and diffraction are two sides of the same coin. The both come into play when there are various close sources and obstructions (having dimensions of the order of the wavelength of light used). As such, they must be kept in mind when designing telescopes, interferometers, diffraction gratings, making precision measurements(such as detection of gravitational waves by LIGO) and has various other applications in modern science.